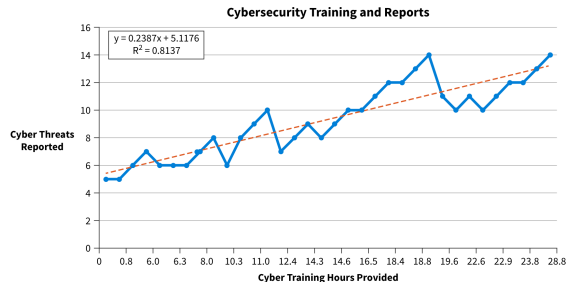


BE 3.9_1 & BE 3.11

Description	Analysis
1. The statistical measures used to evaluate a regression model.	
2. The outcome variable in a regression model.	
3. The variable or variables that influence the outcome variable.	
4. This type of relationship shows steady increases or decreases over the range of the independent variable.	
5. This statistic measures the strength of the relationship between the dependent and independent variables.	
6. This statistic explains how well the regression line fits the data.	



BE 3.12 & BE 3.13

Definition	Term
1. The process of selecting values of variables that minimize or maximize some quantity of interest.	
2. The unknown values a model seeks to determine.	
3. The mathematical equation that describes the output target that we seek to minimize or maximize.	
4. The limitations, requirements, or other restrictions that must be imposed on any solution.	

	Standard	Blue Flying	Red Swimming	Yellow Hopping	Total
Number Produced	1	1	1	1	
CM per unit	17.87	15.20	21.00	19.87	
Total CM	17.87	15.20	21.00	19.87	73.94
Machine Hours Per Unit	6	8	12	11	0
Total Machine Hours	6	8	12	11	37
Constraints					
Machine Hours					50,000
Demand	1,200	1,750	1,670	1,250	

BE 3.13_1 & BE 3.16

	Standard	Blue Flying	Red Swimming	Yellow Hopping	Total
Number Produced	1	1	1	1	
CM per unit	17.87	15.20	21.00	19.87	
Total CM	17.87	15.20	21.00	19.87	73.94
Machine Hours Per Unit	6	8	12	11	0
Total Machine Hours	6	8	12	11	37
Constraints					
Machine Hours					50,000
Demand	1,200	1,750	1,670	1,250	

Statement	Practice Area
1. Simone and the information technology team are investigating the possibility of investing in new technologies.	
2. Chung would like to improve the company's information system performance and is evaluating processing time to prepare month-end management reports.	
3. Christine wants to identify unusual sales transactions that may have contributed to an unexpected increase in the client's revenue compared to last year.	
4. Daniel is currently using the Scenario Manager function in Excel to identify interest rates, payment terms, and length of a loan to evaluate various capital requirement strategies.	
5. Ellis is analyzing a client's tax liability calculation to identify if there are expenses that are not properly included as deductions.	
6. Pierre is currently examining the relationship between machine hours and maintenance expenses to predict future manufacturing modifications.	
7. Mica is performing a regression analysis to estimate the predicted monthly revenue for a retail client. He will then compare the predicted monthly revenue to the client's recorded amount.	

Phần Bài tập

Exercises (EX)

EX 3.1 & EX 3.4

Objective	Initial Question	Sub-questions	Possible Measures
Understand sales by customer for the month of June.	What happened to sales in the month of June?	What was total sales? What was the average sales amount? What was total sales by customer? Which customers had the highest sales?	1. Sum of sales amount (total measure). 2. Average of sales amount (measure of location). 3. Total sales separated by customer. 4. Total sales separated by customer using a highlight table.

Objective	Initial Question	Sub-questions	Possible Measures
Understand cybersecurity training expenses.	What is the total amount of cybersecurity training expense for 2023, 2024 and 2025?	1.	2.
	What is the percentage change of cybersecurity training expenses between 2023 and 2024 and between 2024 and 2025?	3.	4.

EX 3.7A & EX 3.7B

Data Field	Description
SalesOrderNumber	Uniquely assigned sales order number.
CustomerID	Uniquely assigned customer number.
CustomerName	A customers name.
CustomerCredit	Customer's credit limit.
CustomerAddress	U.S Postal mailing address and street name.
CustomerCity	Mailing address city.
CustomerState	Mailing address state.
CustomerZipcode	Mailing address zip.
CustomerPayTerms	Payment terms negotiated with the customer.
SalesOrderDate	Sales order date.
SalesOrderID	Uniquely identifiable item number.
SalesOrderDescription	Description of item.
ItemPrice	Sales price of the item per negotiated terms with the customer.
ItemQuantity	Total quantity of items sold to the customer.
InvoiceNumber	Uniquely assigned invoice number.
InvoiceAmount	The total amount of the invoice, including shipping charges and tax.
PaidAmount	The total amount of the payment received from the customer.
PaidDate	The date the invoice was paid.

Objective	Initial Question	Sub-questions	Possible Measures
Understand why accounts receivable increased compared to the prior year.	1.	2.	3.

EX 3.8 & EX 3.8_1

Objective	Initial Questions	Sub-questions	Possible Measures
Determine why cyber incident reports are increasing.	1.	Are there more cyber incidents reporting during a specific period of time?	Create a line chart of cyber incident reports over time and review for a trend or pattern.
	2.	3.	Perform a correlation between training expenses and cyber incidents reported by using the CORREL function in Excel.

Objective	Initial Questions	Sub-questions	Possible Measures
Determine why cyber incident reports are increasing.	1.	Are there more cyber incidents reporting during a specific period of time?	Create a line chart of cyber incident reports over time and review for a trend or pattern.
	2.	3.	Perform a correlation between training expenses and cyber incidents reported by using the CORREL function in Excel.

EX 3.10 & EX 3.13

Objective	Initial Question	Sub-question	Possible Measures
Predict sales returns.	How much will sales returns change if we have a 10% increase in sales quantity?	1.	Regression analysis

PREDICTED REVENUE MODEL

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.997170819							
R Square	0.994349643							
Adjusted R Square	0.994007197							
Standard Error	27302.53042							
Observations	36							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	2	4,328,953,815,697.83	2,164,476,907,848.92	2,903.67	0.00			
Residual	33	24,599,129,513.81	745,428,167.09					
Total	35	4,353,552,945,211.64						
	Coefficients		Standard Error	t Stat	P value	Lower 95%	Upper 95%	
Intercept	-597355.3555	79141.28598	-7.547961195	1.10769E-08	-758369.5125	-436341.1985	-758369.5125	-436341.1985
Volume	424.0763009	7.525920213	56.34876386	2.20655E-34	408.7647011	439.3879007	408.7647011	439.3879007
Average Price	1453.623412	208.2682902	6.97957145	5.57457E-08	1029.89839	1877.348434	1029.89839	1877.348434

EX 3.13_1 & EX 3.16

PREDICTED REVENUE MODEL

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.997170819							
R Square	0.994349643							
Adjusted R Square	0.994007197							
Standard Error	27302.53042							
Observations	36							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	2	4,328,953,815,697.83	2,164,476,907,848.92	2,903.67	0.00			
Residual	33	24,599,129,513.81	745,428,167.09					
Total	35	4,353,552,945,211.64						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-597355.3555	79141.28598	-7.547961195	1.10769E-08	-758369.5125	-436341.1985	-758369.5125	-436341.1985
Volume	424.0763009	7.525920213	56.34876386	2.20655E-34	408.7647011	439.3879007	408.7647011	439.3879007
Average Price	1453.623412	208.2682902	6.97957145	5.57457E-08	1029.89839	1877.348434	1029.89839	1877.348434

Product	Gross Profit per Item	Crafting Hours per Item
Ball cap	5.00	4
GO Team t-shirt	7.50	2.5
Defense Wins Championships t-shirt	7.50	3
Socks	3.30	6
Knee-high socks	4.50	3.5

EX 3.17

	Necklaces	Bracelets	Earrings
Contribution Margin per unit	\$ 12.00	\$ 8.00	\$ 14.00
Resource Requirements:			
Direct Materials, ounces per unit	22	8	13
Design Hours per unit	4	1	5
Machining Hours per unit	2	1	2

Tình huống Ứng dụng Chuyên môn

Professional Application Cases (PAC)

PAC 3.1 & PAC 3.2

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand gross patient revenue (GrossPatientRevenue) for the year 2019, which is the year under audit.	What happened to gross patient revenue during 2019?	1.	2.
Diagnostic Analysis: Understand what happened to gross patient revenue for the year 2019 compared to 2018, Specifically, GrossPatientRevenue, GrossInpatientRevenue, and GrossOutpatientRevenue data points.	Which locations had changes in revenue from 2018 to 2019?	3.	4.
Predictive Analysis: Develop an expectation for gross patient revenue in 2019 based on the components patient days, out patient visits, and average length of stay.	What factors drive changes in gross revenue?	5.	6.

Variables	Definition per Data Dictionary
HospitalFTE	Number of hospital full time employees
NurseFTE	Average number of nursing personnel
ProductiveHours	Total productive hours
NonProductiveHours	Total nonproductive hours
PaidHours	Total paid hours
MedicalStaff	Number of active medical staff

Complete the chart.

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand labor information for 2018 - 2019.	What happened to labor hours during 2018 and 2019?	1.	2.
Diagnostic Analysis: Determine why variables changed between 2018 and 2019.	Which hospitals have increasing salary expenses from 2018 to 2019?	3.	4.
Predictive Analysis: Identify if there is relationship between total productive hours and gross patient revenue.	Can productive hours predict gross patient revenue?	5.	6.

PAC 3.3 & PAC 3.4

Variables	Definition per Data Dictionary
DayTotal	Patient Days Total
OccupancyLicensed	Occupancy rate – Licensed beds
OccupancyAvailable	Occupancy rate – Available beds
AverageLengthStayAll	Average Length of Stay – Including Long Term Care
AverageLengthStayNoLTC	Average Length of Stay – Excluding Long Term Care
GrossInpatientRevenue	Gross Inpatient Revenue

Complete the chart for descriptive, diagnostic, and predictive analytics.

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand gross inpatient revenue for the year 2019, which is the year under audit.	What happened to gross patient revenue during 2019?	1.	2.
Diagnostic Analysis: Understand what happened to gross inpatient revenue for the year 2019 compared to 2018.	How has revenue changed at the locations?	3.	4.
Predictive Analysis: Identify the utilization measures that are related to in-patient revenue.	Are there any measures correlated to inpatient revenue?	5.	6.

IRS 990 Data:

Variables	Definition per Data Dictionary
Assets	Total asset book value at year end
Income	Net Income
Revenue	Revenue reported on line 12 of Form 990

Kindred 2013 – 2019 Financial Data

Variables	Definition per Data Dictionary
TotalAssets	Total Assets
TotalRevenue	Net patient revenue plus other operating revenue
NetIncome	Net Income

Complete the chart.

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand assets, revenue, and income information for Kindred's non-profit and for-profit hospitals	Which hospitals have the highest income?	1.	2.
Descriptive Analysis: Understand how IRS 990 data for non-profit hospitals compares to the Kindred non-profit hospitals	Are the Kindred Hospitals comparable to the sample of IRS 990 filings.	3.	4.
Diagnostic Analysis: Identify trends in assets, revenue, and income for Kindred hospitals.	5.	6.	7.
Diagnostic Analysis: Compare Kindred data to IRS 990 data.	8.	9.	10.

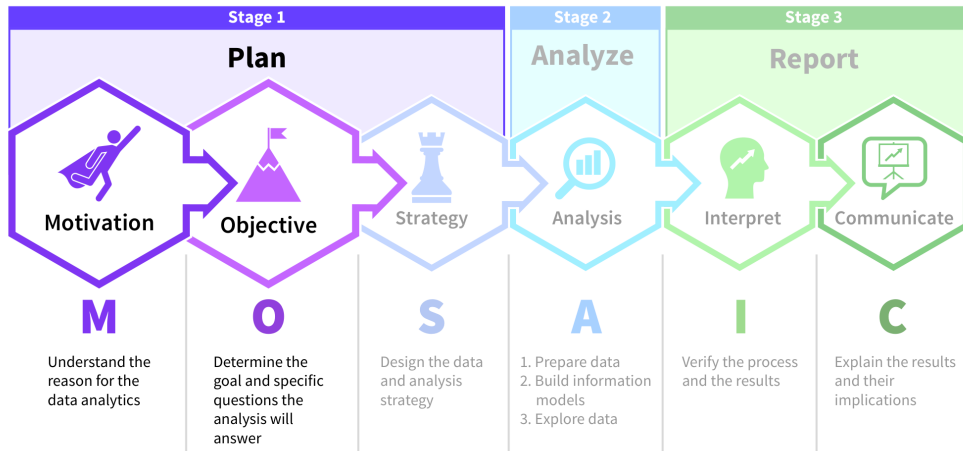
Tài liệu Bổ sung

The
In 20

Healthcare service company 3.1

	2019 (unaudited)	2018 (audited)	2017 (audited)
Gross Patient Revenue	\$2,802,211,999	\$2,515,881,547	\$2,423,431,540
Deductions from Revenue	\$2,206,888,182	\$1,940,233,238	\$1,869,518,824
Net Patient Revenue	\$ 595,323,817	\$ 575,648,309	\$ 553,912,716
Other Operating Revenue	\$ 2,262,385	\$ 2,367,007	\$ 2,663,316
Operating Expenses	\$ 579,928,100	\$ 523,090,033	\$ 506,521,633
Net Revenue from Operations	\$ 17,658,102	\$ 54,925,283	\$ 50,054,399
Non-Operating Revenue	\$ 1,237,718	\$ 970,497	\$ 763,174
Non-Operating Expense	\$ 110,477	\$ 200,367	\$ 385,724
Income Tax	\$ 86,442	\$ 73,200	\$ 25,176
Net Income	\$ 18,698,901	\$ 55,622,213	\$ 50,406,673

Info 3.1



Hỏi & Đáp

Cảm ơn các bạn đã lắng nghe!